Ethical research should be credible research: Do researchers comply with the Declaration of Helsinki?

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Discussion points

- Declaration of Helsinki
- Informative research vs. research waste
- Types of research waste
- Reasons for research waste
- Potential solutions

Declaration of Helsinki: What it is

- The Declaration was developed in 1964 by the World Medical Association
- It serves as the international benchmark for ethical conduct in human research, providing high-level principles that guide regulatory bodies, research institutions, and individual researchers

How many of you do know the content of the Declaration of Helsinki?

How many of you have declared to adhere to the Declaration of Helsinki?

Declaration of Helsinki 2013

12. Medical research involving human subjects must be conducted only by individuals with the appropriate ethics and scientific education, training and qualifications. Research on patients or healthy volunteers requires the supervision of a competent and appropriately qualified physician or other health care professional (World Medical Association 2013)

Declaration of Helsinki 2013

21. Medical research involving human subjects must conform to generally accepted scientific principles, be based on a thorough knowledge of the scientific literature, other relevant sources of information, and adequate laboratory and, as appropriate, animal experimentation. [. . .] (World Medical Association 2013)

Declaration of Helsinki 2024

21. Medical research involving human participants must have a scientifically sound and rigorous design and execution that are likely to produce reliable, valid, and valuable knowledge and avoid **research waste** [. . .] (World Medical Association 2025)

Declaration of Helsinki 2024

36. Researchers, authors, sponsors, editors, and publishers all have ethical obligations with regard to the publication and dissemination of the results of research. Researchers have a duty to make publicly available the results of their research on human participants and are accountable for the timeliness, completeness, and accuracy of their reports. All parties should adhere to accepted guidelines for ethical reporting. Negative and inconclusive as well as positive results must be published or otherwise made publicly available (World Medical Association 2025)

Informative research

- The process of gathering and analyzing data to produce trustworthy knowledge
- It is carried out through ethical and rigorous methods, and communicated transparently and without bias, enabling peers to verify and build upon the findings

Why do we do research?

- To test the the efficacy/effectiveness of interventions
- Is intervention A more effective than intervention B in reducing time to return-to-play in football players after injury?

Research waste

Research that fails to produce reliable or relevant results that cannot be be used to inform decision making

Research waste



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LONDON, SATURDAY 29 JANUARY 1994

The scandal of poor medical research

We need less research, better research, and research done for the right reasons

(Altman 1994)

Waste at four stages of research

Question -> Design and analysis -> Reporting -> Publication

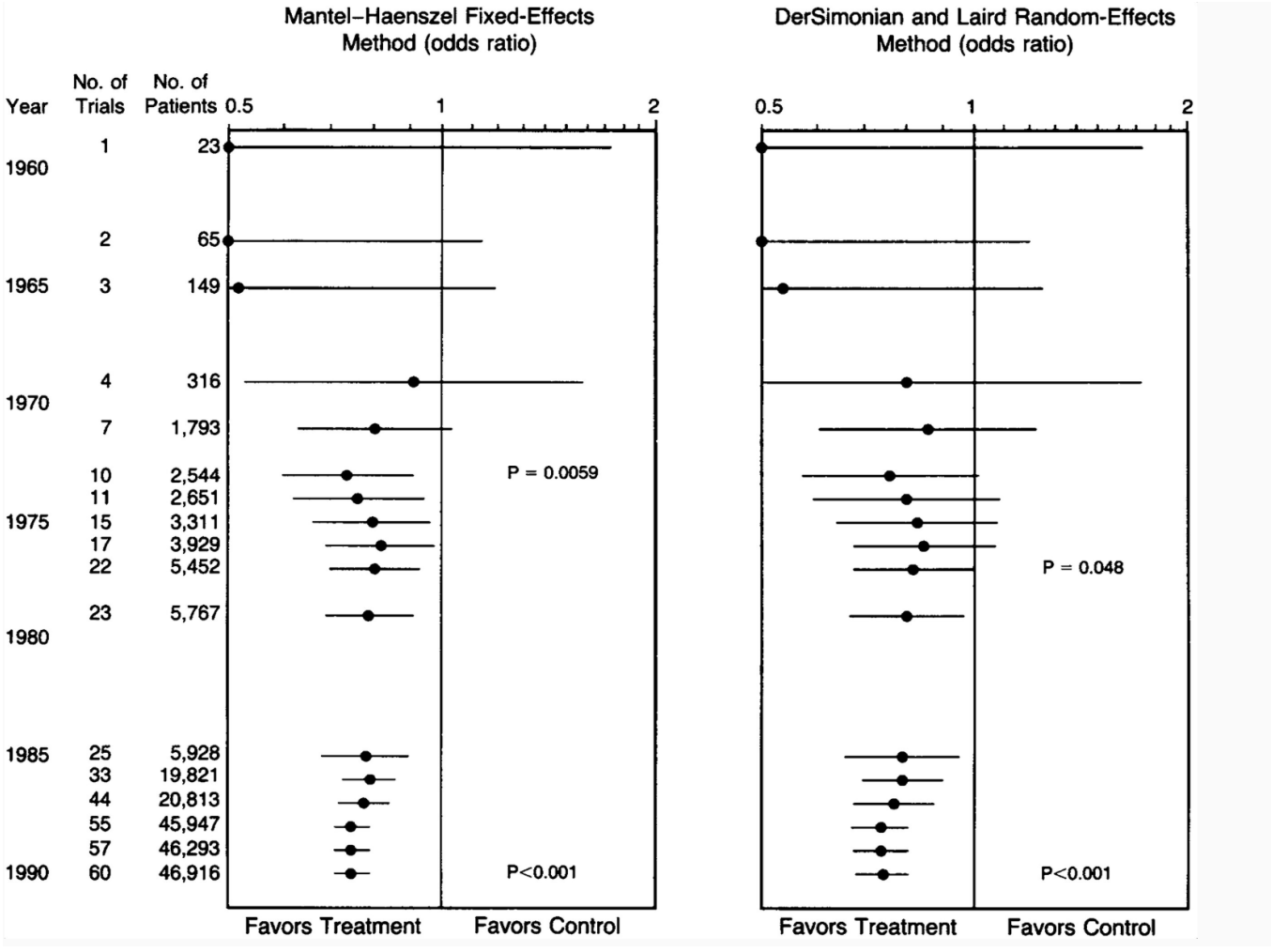
1-Research question

- Irrelevant/low priority questions addressed
- Redundancy

Example: Cumulative Meta-Analysis of Therapeutic Trials for Myocardial Infarction (Lau et al. 1992)

- Thirty-three trials evaluating this therapy were performed between 1959 and 1988
- Statistically significant reduction in mortality (odds ratio, 0.74; 95% CI [0.59 to 0.92]) was achieved in 1973, after only eight trials involving 2432 patients
- The results of the 25 subsequent trials (34,542 additional patients) through 1988 had little or no effect on the odds ratio establishing efficacy, but simply narrowed the 95% CI
- Two very large trials in 1986 (11,712 patients) and in 1988 (17,187 patients) did not modify the already established evidence of efficacy

Cumulative Meta-Analysis of Therapeutic Trials for Myocardial Infarction



(Lau et al. 1992)

Example in sports and exercise science: The effect of self-talk interventions on sports/motor performance

- Since Hatzigeorgiadis et al. (2011) meta-analysis the number of studies conducted examining the effects of self-talk interventions on sport/motor performance has roughly doubled
- 35 new studies published from November 2011 up to November 2023
- The overall effect in Hatzigeorgiadis et al. (2011) was 0.48 95% CI (0.38; 0.58) and after adding newer studies the overall effect was 0.47 95% CI (0.39; 0.56) in Corcoran and Steele (2024)

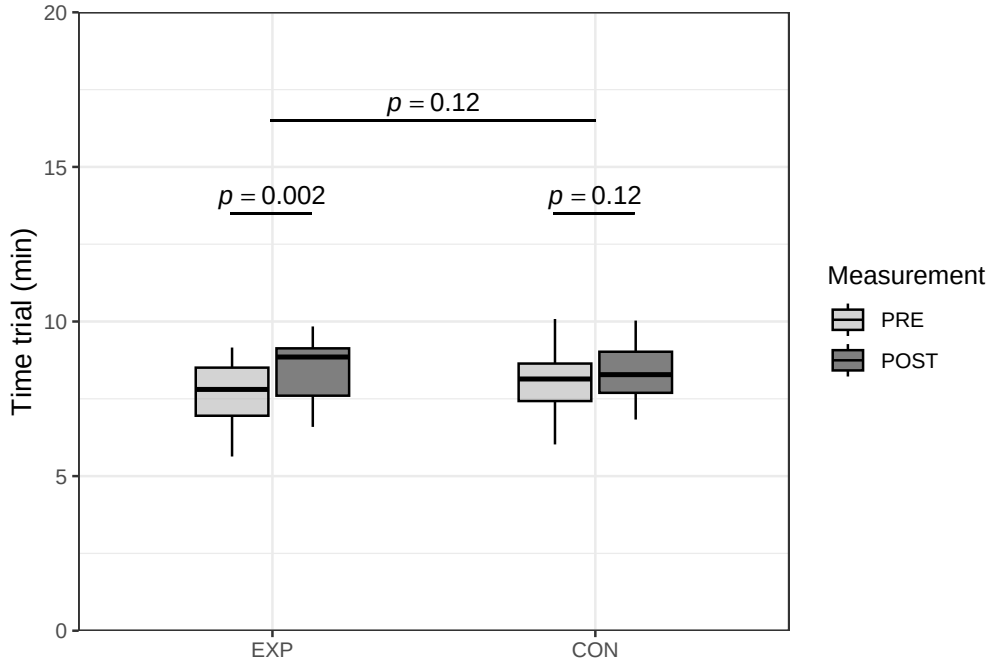
Relevant research question

New research should be set in the context of an evidence gap and should involve stakeholders when possible

2- Study design

Conduct studies that do not answer the research question (Mesquida et al. 2025d). For instance:

- 23 out of 350 studies tested a hypothesis of equivalence using a classic null hypothesis test (Mesquida et al. 2025a)
- Testing for treatment effects by separate analyses of changes from baseline in each group is a misleading approach

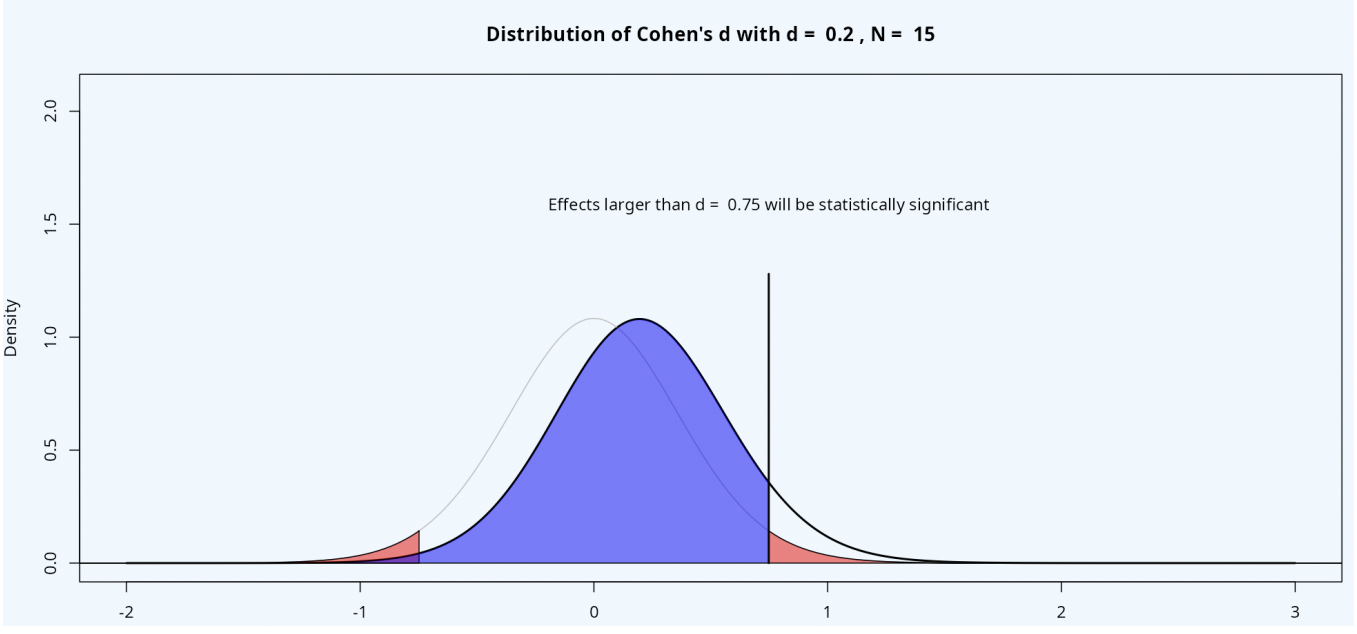


2- Study design

Invalid a priori power analyses (Mesquida et al. 2025b):

- Mean sample size of $\sim$ 30 participants
- 40% (139 out of 350) of hypothesis-testing studies reported conducting an a priori power analysis.
- No significant difference in sample size between studies that reported an a priori power analyses and those that did not
- No significant difference in the proportion of hypotheses supported between studies that reported an a priori power analyses and those that did not
- Potential reasons: sample size bamba (Schulz and Grimes 2005), invalid effect sizes, inflated effect sizes

How a power analysis implicitly reveals the smallest effect size you care about



2- Study design

- **Low statistical power**: an analysis of 269 hypothesis-testing studies showed that the average statistical power was 11% (Mesquida et al. 2025a)
- **Implication**: With such low power, the chance of detecting true effects is extremely low
- Flipping a coin (50% probability).

Why designing studies with high statistical power is important?

1- Interpretation of null findings

- If a study is designed with low power (e.g., 20%), a null significant result cannot be interpreted because it could be due to that the effect size is smaller than expected or non-existent.

Why designing studies with high statistical power is important?

2- Narrower confidence intervals

- Example 1: study with 20 participants yields a wider 95% CI [-0.72; 1.16]
- Example 2: study with 200 participants yields a narrower 95% CI [-0.75; -0.23]

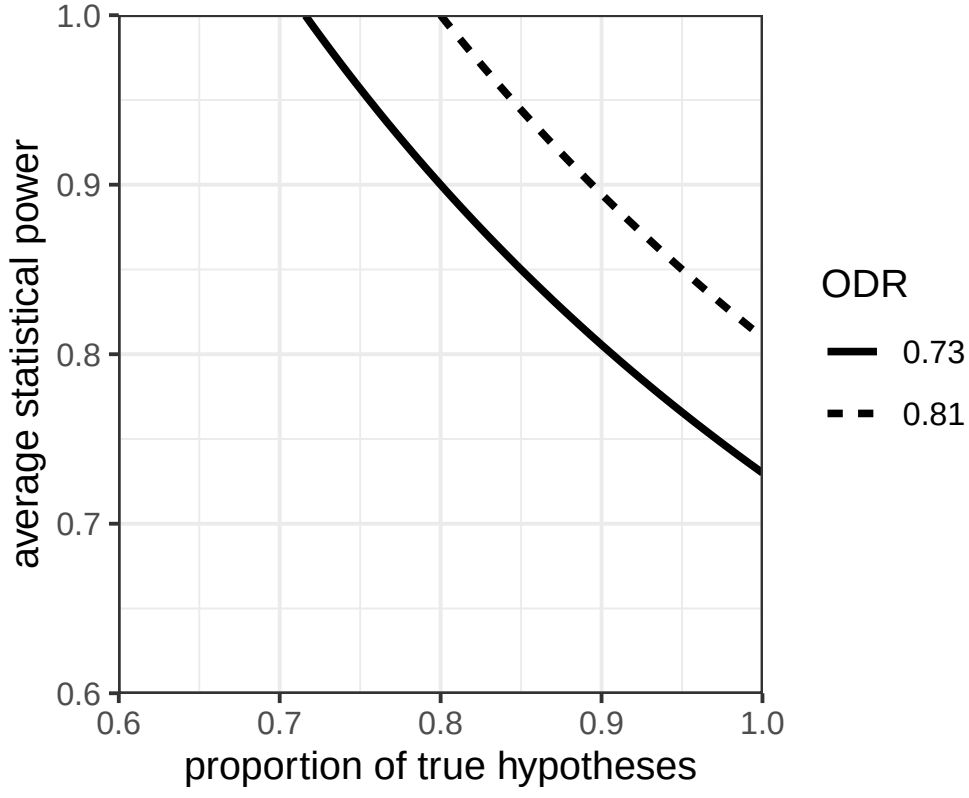
2- Study design

Lack of transparency

- Framing your study as confirmatory when is exploratory (Ditroilo et al. 2025)
- Vague and unfalsifiable hypotheses (Mesquida et al. 2025d)

3- Selection bias

- Publication bias
- Flexibility in data analysis



(Mesquida et al. 2025a)

Publication bias

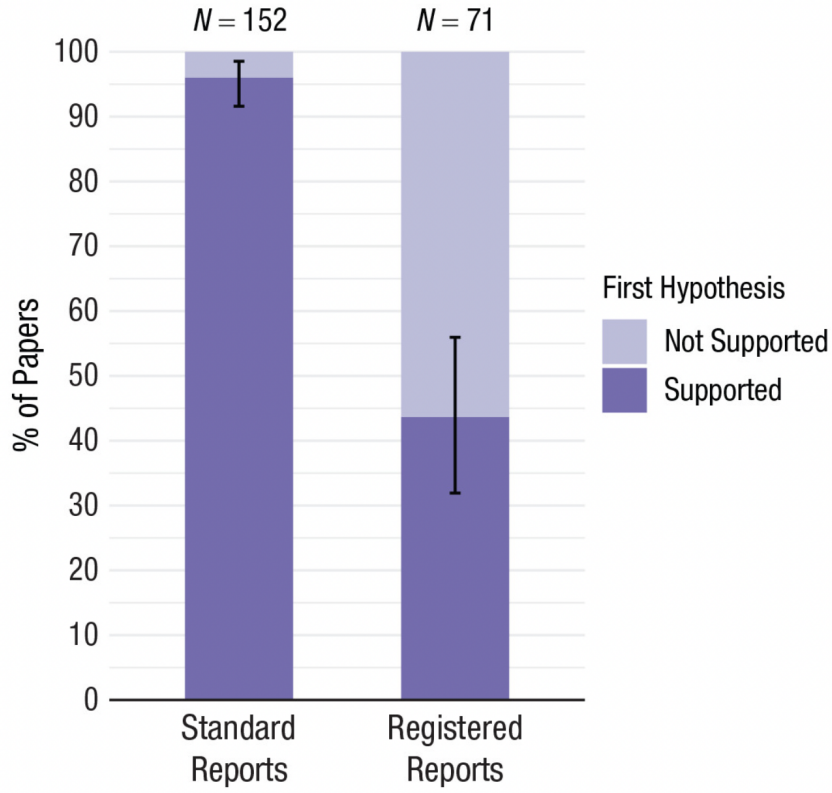


Fig. 2. Positive result rates for standard reports and Registered Reports. Error bars indicate 95% confidence intervals around the observed positive result rate.

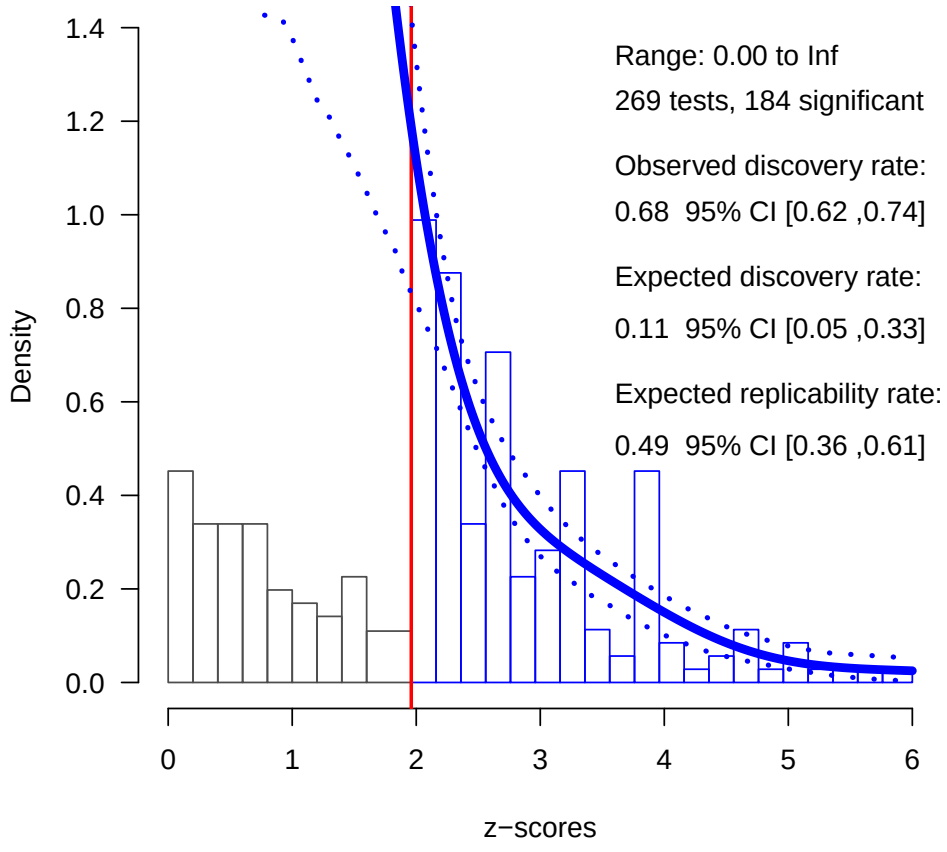
(Scheel et al. 2021)

Flexibility in data analysis



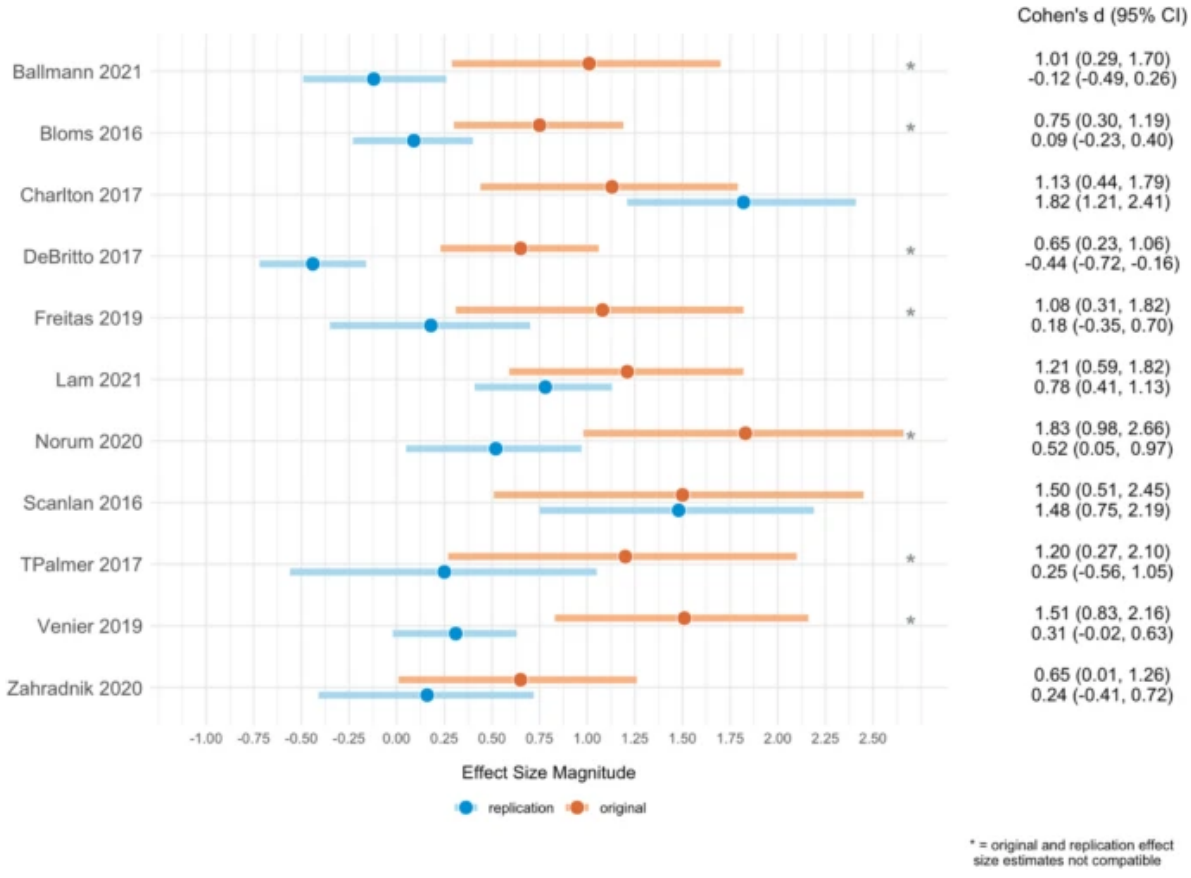
Figure 1: Distribution of effect sizes under the null hypothesis when performing 5 indepdendent t-tests, the correlation between dependent variables is 0.5 and alpha level is 0.05

Selection bias and statistical power in sports and exercise science



(Mesquida et al. 2025a)

Consequences of selection bias and studies with underpowered designs



(Murphy, Caldwell, Mesquida, et al. 2025)

Results of the Sports Science Replication Project (Murphy, Caldwell, Mesquida, et al. 2025)

- Inflated type I error rate: 56% of the 25 analysed replication studies had a statistically significant p-value similar to the original studies
- Inflated effect sizes: median percentage decrease in magnitude from the original to the replication effect size estimate equal to 75% for all of the replication effect sizes
- Only 36% of replication and original effect size estimates were compatible

Results from other meta-scientific studies

Study	Topic	Original ES	Adjusted-bias ES
McKay, Yantha, et al. (2022)	Effect of self-control on motor learning	$g = 0.44$	$g = 0.11$
Ciria et al. (2023)	Effect of physical exercise on cognition	$d = 0.22$	$d = 0.05$
McKay et al. (2024)	Effect of external focus on the intended movement effect	$g = 0.26$	$g = 0.01$
Holgado et al. (2023)	Effect of mental fatigue on exercise performance	$d_z = 0.54$	$d_z = 0.01$

ES = standardized effect size

4- Reporting of statistical information and data sharing

- "This project faced many practical challenges when preparing studies for replication, specifically the poor reporting of statistical information, the availability of original raw data [. . .]" (Murphy, Caldwell, and Warne 2025)
- Out of 350 hypothesis-testing hypotheses, 57% (46/81) could not be recomputed into an exact p-value due to poor reporting of statistical information or availability of original raw data (Mesquida et al. 2025a)

4- Reporting of statistics and data sharing

Discipline/sub-discipline	All eligible Q1 'sports science' journal articles (*n* = 299)	Sport and exercise science † (*n* = 148/299)	Sports medicine (*n* = 137/299)	Rehabilitation (*n* = 14/299)
Sharing data	13/299	8/148	5/137	0/14
	4.3% [2.3–7.3]	5.4% [2.4–10.4]	3.6% [1.2–8.3]	0.0%
Sharing code	0/298	0/148	0/136	0/14
	0.0%	0.0%	0.0%	0.0%
Data available upon request ‡	5/286	2/140	3/132	0/14
	1.7% [0.6–4.0]	1.4% [0.2–5.1]	2.3% [0.5–6.5]	0.0%

(Borg et al. 2020)

Why sharing data and code are important?

- Verify results: Allow peers to reproduce and confirm the findings
- Reuse data: Facilitate meta-analyses, replications, secondary analyses, or new research projects
- Increase transparency: Promote trust and accountability in research

Research waste also affect meta-analyses

In the 30 most cited meta-analyses in S&C research, Kadlec et al. (2023) found that:

- Almost 45% contained at least one effect size that was mistakenly calculated using SE rather than SD.
- 45% failed to account for effect size dependency

In 96 meta-analyses published in Sports Medicine, Mesquida et al. (2025c) found that:

- 16% of 96 meta-analyses contained data to reproduce study effect sizes
- Many combined studies with different designs (e.g., within- and between-designs, RCT and non-RCT)
- 34% did not account for effect size dependency

Summary

- Research waste in the form of studies that do not answer the research question, inflated effect sizes, study designs with low power, poor reporting of statistical practices and lack of data sharing and low methodligical quality of meta-analyses
- We, as a community, are not properly trained to conduct informative research

Drivers of research waste

Wrong incentives:

- Quantity vs. quality: mass produce sports and exercise science research for career purposes
- Preference for studies that report positive findings
- Compete for limited resources

Drivers of research waste

- Time constraints: lack of time to develop expertise and skills
- Incompetence: insufficient training

How to avoid research waste



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Long-term solutions

- Reward researchers not for the number of publications but for the quality of their research
- More time for improving research competences
- Improve statistical training at BSc and MSc

How to avoid research waste

Mid-term solutions

- Science as a collaborative effort where each member brings specialized expertise
- Big Team Science

Question -> Design -> Analysis Plan -> Data collection -> Analysis execution -> Reporting -> Interpretation -> Data stewardship

Science as a collaborative effort

- The goal is not to conduct experiments alone or with small group of collaborators which may lack expertise in some areas
- The goal is to support and ehance collective scientific research

Science as a collaborative effort

- Theorists -> develop frameworks or hypotheses to explain phenomena
- Methodologists -> statisticians ->design studies and slect appropiate methods
- Data collectors -> gather and organize raw data
- Experts in interpreting effect sizes -> assess the practical significance of findings
- Data stewards -> manage, curate and maintain datasets to ensure FAIR principles and reproducibility

Specialization

- **12**. Medical research involving human participants must be conducted only by individuals with the appropriate ethics and scientific education, training and qualifications. Such research requires the supervision of a competent and appropriately qualified physician or other researcher (World Medical Association 2025)

How to avoid research waste

Short-term solutions

- Collaboration with applied statisticians (Sainani et al. 2021)
- Registered Reports (?)

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